

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 11: Transportation in Animals and Plants

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	11 — Transportation in Animals and Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. At this hardest tier every classic flip — arteries-low-pressure, veins-have-valves-because-pressure-is-high, atria-pump/ventricles-receive, pulse-is-a-valve-snap, urea-is-stored-food or made-in-the-bladder, xylem-carries-food, phloem-only-downward, transpiration-stops-in-heat, wilting-means-poisoned-soil, blood-is-blue, diffusion-suffices-for-large-bodies — is buried inside chained rate and ratio computations, multi-vessel and multi-organ traces, and animal-plant cross-system synthesis, so a half-remembered rule still lands on a plausible wrong option. Do not write on this paper — mark answers on your answer sheet.

1. A patient's chart lists three independent results: oxygen reaches the tissues normally, a finger-prick stops bleeding in the usual time, but a small skin wound becomes infected and the infection spreads unusually fast. Reasoning from which single function has failed while the other two are intact, the most plausibly deficient component, and the reason, is:

- (A) Red blood cells, because a spreading skin infection is really a sign that too little oxygen is reaching the wound
- (B) Platelets, because an infection that spreads means the wound never truly sealed with a clot
- (C) White blood cells, because germ defence is the only listed function failing while oxygen carriage and clotting are normal
- (D) Plasma, because the watery part of the blood is what engulfs and destroys invading bacteria

2. Three centrifuged blood tubes settle into a clear pale-yellow upper column over a packed red layer. Tube P: 55 percent yellow, 45 percent packed. Tube Q: 60 percent yellow,

40 percent packed. Tube R: 40 percent yellow, 60 percent packed. Ranking the three by the proportion of cells (mostly red cells) from most to least, the order is:

- (A) Q, then P, then R, because the tall yellow column in Q is where the red cells collect
- (B) P, then Q, then R, because the tube with the most balanced split always carries the most cells
- (C) Q, then R, then P, because a thicker yellow layer always means more haemoglobin and so more cells
- (D) R, then P, then Q, because the packed lower layer is the cells and R has the largest packed fraction

3. A mature human red cell has lost its nucleus and is shaped to maximise haemoglobin packing, while a white cell keeps its nucleus. A student must explain why this difference suits each cell's job. The soundest explanation is:

- (A) The red cell keeps no nucleus so it can divide faster than the white cell to replace lost blood quickly
- (B) The white cell drops oxygen carriage and keeps a nucleus only so it can clot the blood at wounds
- (C) Both cells would carry more oxygen if the white cell also lost its nucleus, since nuclei block haemoglobin everywhere
- (D) Losing the nucleus frees internal room for more haemoglobin so the red cell carries more oxygen, while the white cell keeps its nucleus because defending against germs needs the machinery a nucleus directs

4. A clinic sees two patients. Patient 1 tires quickly and is breathless after mild effort; iron in the diet is low. Patient 2 bleeds for a long time from tiny cuts. A trainee must map each patient to the cause and to the dietary or cellular fix. The fully correct mapping is:

- (A) Patient 1 lacks platelets so oxygen delivery is slow; Patient 2 lacks haemoglobin so wounds stay open
- (B) Patient 1 lacks white cells causing breathlessness; Patient 2 lacks white cells causing slow clotting
- (C) Patient 1 has too much plasma diluting oxygen; Patient 2 has too many red cells thickening the clot
- (D) Patient 1 lacks haemoglobin (helped by dietary iron) so tissues get little oxygen; Patient 2 lacks platelets so clots form slowly

5. A nutrition chart claims that eating iron-rich food makes blood 'stronger.' To justify this strictly by function, a student must connect iron to a measurable outcome. The justification that holds up is:

- (A) Iron is built into haemoglobin, so adequate iron lets each red cell carry more oxygen to the tissues
- (B) Iron becomes the platelets, so without enough iron the blood loses its ability to clot at wounds

- (C) Iron forms the liquid plasma, so without iron there is no fluid for the blood cells to float in
- (D) Iron is what the white cells are made of, so without iron the body cannot fight off germs

6. During an infection a person's white-cell count climbs steeply, peaks, then settles back to normal as they recover, while their red-cell count barely changes throughout. The pattern is best explained by:

- (A) Red cells convert into white cells during the illness and convert back into red cells on recovery
- (B) Platelets multiply into white cells to clot the infected blood and then dissolve as the fever breaks
- (C) Plasma thickens into white cells during the fever and thins back into liquid afterwards
- (D) The body makes extra defenders while germs are present and reduces them once the threat clears, so white cells rise and fall while red cells stay steady

7. Plasma is the liquid that carries the blood cells, but it also ferries dissolved cargo. A student lists what plasma realistically transports in a Class-7 picture. The correct list is:

- (A) Oxygen as its main cargo, because plasma is the part of blood that binds and carries oxygen
- (B) Only the solid clot, which plasma carries whole to the next wound that needs sealing
- (C) Digested food, the waste urea, and other dissolved substances, all carried dissolved in the liquid plasma
- (D) Whole undigested food pieces and bone fragments that plasma ships to repair sites

8. After heavy blood loss a patient is given a transfusion specifically to restore oxygen delivery to the tissues, not to fight infection or to stop bleeding. The component of donated blood that most directly meets that stated goal, and why, is:

- (A) Red cells, because their haemoglobin is what carries oxygen to the tissues
- (B) Platelets, because oxygen reaches tissues only after the blood has clotted quickly enough
- (C) White cells, because fighting germs is the route by which oxygen is delivered to tissues
- (D) Plasma proteins, because the liquid part of blood is what binds and carries the oxygen

9. From the same healthy person, blood drawn from the vessel leaving the lungs is bright cherry-red, while blood drawn from the vessel entering the lungs is darker, duller red. Reading both samples, the brightest sample and the reason are:

- (A) The sample entering the lungs is brighter because blood collects platelets as it travels back toward the lungs
- (B) The sample leaving the lungs is brighter because it is carrying more oxygen bound to its haemoglobin after gas exchange
- (C) The sample leaving the lungs is brighter because lung blood is the only blood that is genuinely red rather than blue
- (D) Both samples are equally bright, since colour cannot change as blood passes through the lungs

10. Under a microscope a single drop of blood shows one cell type vastly outnumbering all others; it is small, round, biconcave and lacks a nucleus. A second, far rarer type is larger and keeps a nucleus. Matching each described type to its job gives:

- (A) The most numerous nucleus-free cells are white cells that clot the blood; the rarer cells are red cells that carry oxygen
- (B) The most numerous nucleus-free cells are red cells that carry oxygen; the rarer nucleated cells are white cells that fight germs
- (C) The most numerous cells are platelets that carry oxygen; the rarer cells are red cells that fight germs
- (D) The most numerous nucleus-free cells are red cells that fight germs; the rarer cells are white cells that carry oxygen

11. A model heart-and-blood kit lists four parts to label: a thick stiff pipe, a floppy valved return pipe, a fine one-cell-thick mesh, and a red liquid full of nucleus-free discs and rare nucleated cells. Matching the liquid's contents to function, the nucleus-free discs and the rare nucleated cells respectively do:

- (A) The nucleus-free discs fight germs; the rare nucleated cells carry oxygen through the thick pipe
- (B) The nucleus-free discs clot the blood; the rare nucleated cells form the liquid plasma
- (C) The nucleus-free discs carry oxygen; the rare nucleated cells fight germs
- (D) The nucleus-free discs carry urea to the kidney; the rare nucleated cells carry oxygen to the mesh

12. In an accident both an artery and a vein are severed in the same limb. The artery throws blood out in strong spurts timed with the heartbeat; the vein bleeds in a steady low ooze. Synthesising the heart's action with vessel structure, the spurting artery behaves so because it:

- (A) Carries blood pumped straight from the heart at high pressure that rises and falls with each beat, so it spurts in time with the beat
- (B) Carries blood slowly back toward the heart, and that slow return is what makes it surge out in bursts

(C) Is lined with many valves that snap open and shut each beat, and those snaps produce the spurts

(D) Has a thinner wall than the vein, so blood bursts out of it more forcefully than from the vein

13. A student measures four vessels. V1: thick muscular wall, no valves, blood pulses out. V2: thin wall, valves present, low steady flow. V3: wall one cell thick, no valves, slowest flow. V4: thick muscular wall, no valves, but carries blood toward the lungs. Naming all four:

(A) V1 and V4 are arteries, V2 is a vein, V3 is a capillary

(B) V1 is an artery, V2 is a capillary, V3 is a vein, V4 is a vein

(C) V1 is a vein, V2 is an artery, V3 is a capillary, V4 is an artery

(D) V1 and V2 are arteries, V3 is a vein, V4 is a capillary

14. A classmate writes: 'Veins carry valves because the blood inside them is at such high pressure that valves are needed to brake it.' Correcting both the pressure claim and the purpose of the valves, the accurate statement is:

(A) Vein blood is indeed at high pressure, and the valves brake it exactly as the classmate states

(B) Vein blood is at low pressure, and the valves stop it slipping backward so it keeps heading to the heart

(C) Veins have valves only to make the heartbeat sound louder through a stethoscope

(D) The valves in veins actively pump the blood forward the way the heart's muscle does

15. Trace a single red cell by vessel TYPE on a complete trip: it leaves the heart, gives up oxygen in a toe muscle, then returns to the heart. The correct ordered sequence of vessel types is:

(A) Vein, then capillaries, then artery (B) Artery, then capillaries, then vein

(C) Capillaries, then artery, then vein (D) Artery, then vein, then capillaries

16. A student insists every artery carries oxygen-rich blood and every vein carries oxygen-poor blood, with no exceptions anywhere. The most accurate correction names the exception precisely:

(A) It is completely true; there is no place in the body where an artery carries oxygen-poor blood

(B) It is usually true, but the artery going to the lungs carries oxygen-poor blood and the vein returning from the lungs carries oxygen-rich blood

(C) It is completely false; arteries always carry oxygen-poor blood and veins always carry oxygen-rich blood

(D) It holds only in adults and reverses entirely once a person is a child

17. Comparing the whole circulation, a student must say where pressure is HIGHEST and where flow is SLOWEST and link both to function. The correct pairing is:

- (A) Pressure is highest in the veins, and flow is slowest in the arteries, so the heart works least there
- (B) Pressure is highest in the capillaries, and flow is slowest in the veins, which is why veins need valves
- (C) Pressure is highest in the arteries, and flow is slowest in the capillaries, which gives time for exchange with body cells
- (D) Pressure is highest and flow is slowest both in the arteries, which is why they pulse

18. Through pale forearm skin, vessels look bluish, and a child concludes the blood inside is literally blue. Combining what colour blood actually is with why it looks bluish, the correct account is:

- (A) The blood truly is blue inside the vein and turns red only when it meets air at a cut
- (B) The blood is dark red; it only looks bluish because of how light passes through the skin and vessel walls
- (C) Only the oxygen-poor blood in those vessels is genuinely coloured blue inside the body
- (D) The bluish look comes from blue-coloured plasma filling those particular vessels

19. A capillary is one cell thick and threads between body cells so substances can pass. In a thought experiment its wall is rebuilt thick and muscular like an artery, with everything else unchanged. The most direct consequence is:

- (A) Exchange of gases and substances between blood and body cells would be hindered, because the thin wall is what lets them pass
- (B) Exchange would speed up, because a thicker muscular wall pushes substances out to the cells faster
- (C) The rebuilt vessel would begin pumping blood on its own like a tiny extra heart
- (D) Blood inside would immediately clot, because thick walls always trigger clotting

20. A vessel is described three ways at once: it carries blood toward the heart, it has valves along its length, and when cut it gives a steady low-pressure ooze. Naming the vessel and explaining the valves in one statement gives:

- (A) A vein, whose valves keep its low-pressure blood from slipping backward as it returns to the heart
- (B) An artery, whose valves help it pump blood out at high pressure to the body
- (C) A capillary, whose valves slow exchange across its single-cell-thick wall
- (D) A vein, whose valves are built to withstand the very high pressure inside it

21. The four-chambered heart keeps oxygen-rich and oxygen-poor blood on separate sides. A student must say what advantage this gives the body's cells. The best answer is:

- (A) Keeping them separate mainly makes the lub-dub heartbeat louder through a stethoscope
- (B) Separation lets the blood change colour faster as it sweeps through the chambers
- (C) Keeping the two kinds unmixed lets fully oxygenated blood be sent out, so tissues get the most oxygen per unit of blood
- (D) Separation allows the heart to keep beating without using any muscle at all

22. A student notes the two lower heart chambers have markedly thicker muscular walls than the two upper chambers, and that the upper chambers connect to incoming veins while the lower connect to outgoing arteries. Reading structure as a clue to function, the thicker lower chambers:

- (A) Gently receive blood arriving from the veins, while the thinner upper chambers do the forceful pumping out
- (B) Store urine briefly before it is passed out of the body through the urethra
- (C) Hold the valves whose snapping produces the pulse felt at the wrist
- (D) Pump blood out of the heart with greater force, while the thinner upper chambers mainly receive incoming blood

23. A doctor with a stethoscope hears a repeating lub-dub. A second listener feels a throb at the patient's wrist at the same rate. Linking each sound and the throb to its true source shows that the lub-dub and the wrist throb are produced, respectively, by:

- (A) The lub-dub by blood squeezing through capillaries, and the wrist throb by vein valves snapping shut
- (B) The lub-dub by the lungs filling and emptying, and the wrist throb by blood clotting at the skin
- (C) The lub-dub by the heart's valves closing as blood is pumped, and the wrist throb by an artery expanding as each surge of blood enters it
- (D) Both the lub-dub and the wrist throb by the heart's valves snapping at the wrist itself

24. A throb felt at the wrist matches the heartbeat one-for-one. A student must identify what the throb physically IS, not just that it matches. The pulse is:

- (A) Vein valves snapping shut in rhythm with each heartbeat
- (B) The blood briefly clotting at the skin once per beat and then re-dissolving
- (C) Air being pushed down the arm by the lungs with each breath
- (D) The expansion of an artery each time the heart drives a surge of blood into it

25. A nurse counts 33 pulse beats in 22 seconds for a resting child. Computing the rate per minute and then judging it against a typical resting value of about 72 to 80, the result is:

- (A) 90 beats per minute, which is somewhat above a typical resting rate
- (B) 66 beats per minute, which is somewhat below a typical resting rate

- (C) 33 beats per minute, which is far below a typical resting rate
- (D) 110 beats per minute, which is far above a typical resting rate

26. An athlete's resting heart rate is 70 beats per minute; after a sprint it rises to 140 beats per minute. Comparing a 30-second window in each state and linking the count to what the muscles need, in the post-sprint window the heart pumps:

- (A) 70 times versus 35 times at rest — twice as often — delivering oxygen and food faster and clearing waste quicker for the working muscles
- (B) 35 times versus 70 times at rest — half as often — so the muscles can cool down after effort
- (C) 70 times in both states, since the rate per minute changes but the actual flow does not
- (D) 140 times versus 70 times, because a 30-second window holds a full minute's worth of beats

27. Two resting adults are measured: one at 72 beats per minute and one at 126. Given a typical resting rate near 72 to 80, and recalling the pulse equals the heartbeat, the second adult's reading is best described as:

- (A) Exactly the normal resting rate, since any value above 100 is the standard adult resting figure
- (B) Lower than the usual resting rate, because higher numbers mean a slower, calmer heart
- (C) Much higher than the usual resting rate, since 126 is far above the typical 72 to 80 band
- (D) Impossible for any living person, because no human heart can exceed 100 beats per minute

28. A heart-rate trace from a resting patient shows one beat every 0.75 seconds, evenly spaced. Computing the rate per minute from this interval gives a value closest to:

- (A) About 75 beats per minute, by reading the 0.75 directly as the rate
- (B) About 45 beats per minute, by treating one beat per 0.75 seconds as slow
- (C) About 133 beats per minute, by dividing 100 by 0.75
- (D) About 80 beats per minute

29. Two children are timed. Child A: 20 beats in 15 seconds. Child B: 26 beats in 20 seconds. Converting both to beats per minute and comparing, the correct conclusion is:

- (A) A is 80 and B is 78, so B's heart is beating slightly faster because B had more beats counted
- (B) Both are 78 per minute, since counting longer always gives the same rate
- (C) A is 75 and B is 90, so B is much faster, because longer counts always give higher rates
- (D) A is 80 per minute and B is 78 per minute, so A's heart is beating slightly faster

30. Following one complete heartbeat, oxygen-poor blood from the body is sent to the lungs and oxygen-rich blood from the lungs is sent on to the body, the two streams never mixing. Reading this double-loop arrangement, its central benefit is:

- (A) The body receives oxygen-poor blood directly without it ever having to visit the lungs
- (B) Blood is freshly loaded with oxygen at the lungs before each trip to the body, so tissues always receive oxygen-rich blood
- (C) The heart can skip pumping to the lungs once the blood's oxygen is high enough
- (D) Oxygen-rich and oxygen-poor blood are deliberately mixed to give the body a balanced average

31. A teacher says the heart's job in an animal is matched, in a tall plant raising water, by a different mechanism that needs no pump. The most reasonable matching, and its limit, is:

- (A) The transpiration pull draws water up the xylem, doing the heart-like lifting job although the plant has no actual pump
- (B) A tiny heart hidden in the roots pumps water upward just as the animal heart pumps blood
- (C) The phloem squeezes water up to the leaves under pressure, acting as the plant's pump
- (D) The stomata suck water directly out of the soil, replacing the need for any vessels

32. A student lists four body activities and must pick the one that is NOT a form of excretion. The activity that is not excretion is:

- (A) Removing the waste urea from the blood as urine
- (B) Taking in and digesting food in the gut
- (C) Losing excess water and salts through sweat
- (D) Passing out dissolved nitrogen waste through the kidneys

33. Inside each kidney sit about a million microscopic filtering units that clean the blood, while the bladder merely stores the urine they help make. Naming the units and stating exactly where they sit gives:

- (A) Nephrons, located inside the urinary bladder where the urine is stored
- (B) Stomata, located inside the kidneys, doing the blood filtering
- (C) Nephrons, located inside the kidneys
- (D) Platelets, located inside the kidneys, where they filter the blood

34. A student must arrange the urinary structures in the exact order newly formed urine travels through them on its way out of the body. The correct order is:

- (A) Kidney, then urethra, then bladder, then ureter
- (B) Kidney, then ureter, then urinary bladder, then urethra
- (C) Bladder, then ureter, then kidney, then urethra
- (D) Ureter, then kidney, then urethra, then bladder

35. Trace the waste urea correctly all the way from where it is produced to where it finally leaves the body. The correct trace is:

- (A) Produced in the kidneys, carried by the xylem, and passed out through the stomata
- (B) Produced in the bladder, carried by the arteries, and breathed out by the lungs
- (C) Produced in the body cells, carried dissolved in the blood, filtered out by the kidneys, and passed out in urine
- (D) Produced in the urine itself, carried by the ureters, and stored in the heart

36. A student claims urea should be retained because it is a useful stored food the body draws on when energy runs low. Evaluating this claim correctly:

- (A) It is right; urea is a stored food broken down for energy when supplies run low
- (B) It is wrong; urea is a toxic nitrogen waste that would poison the body if allowed to build up
- (C) It is partly right; urea is kept in the kidneys as an emergency reserve
- (D) It is right; urea is retained because it is needed to keep the blood red

37. On a very hot, active day a footballer sweats heavily and passes little urine; on a cool, restful day she sweats little and passes more urine. Reading the kidneys and sweat glands together, this pattern shows they are:

- (A) Working against each other, so heavy sweating forces the kidneys to stop filtering the blood entirely
- (B) Doing wholly unrelated jobs, since sweat only cools the body and never removes any waste
- (C) Both absorbing oxygen through the skin to replace what is lost in heavy sweating
- (D) Sharing the job of removing water and wastes, so when sweating rises the kidneys can let urine fall, and vice versa

38. When both kidneys fail, harmful wastes pile up in the blood and a machine is used to clean it. Naming the treatment and stating exactly what organ's job it replaces:

- (A) Dialysis, which takes over the kidneys' job of filtering wastes from the blood
- (B) Transfusion, a machine treatment that takes over the kidneys' filtering job
- (C) Transpiration, which is the name for a machine that cleans human blood
- (D) Circulation, a machine treatment that replaces the heart's pumping when kidneys fail

39. Plants have no kidneys yet still deal with their wastes. A student must pick the correct description of plant waste handling. The correct description is:

- (A) Plants filter all their waste through a pair of plant kidneys connected to ureters
- (B) Plants send every waste up the xylem and out through the root hairs into the soil
- (C) Plants release some waste through stomata and store some in parts like leaves, bark, gums and resins
- (D) Plants make no waste of any kind and so need no waste-removal system at all

40. Sticky gum oozing from a tree trunk, and salts that accumulate in older leaves which later fall, are both best interpreted in the same way. That interpretation is:

- (A) They are food the plant is deliberately sending down to its roots for storage
- (B) They are water being pulled upward through the trunk toward the leaves
- (C) They are oxygen the plant releases through its bark during the night
- (D) They are ways a plant stores or releases its waste products

41. Comparing plant excretion with animal excretion, a student must pick the most accurate Class-7 statement. That statement is:

- (A) Plant excretion is generally slower and simpler, since plants make less harmful waste and can store some of it
- (B) Plant excretion is faster than animal excretion, because plants have many small kidneys working at once
- (C) Plant excretion is carried out by a heart pumping waste through dedicated waste vessels
- (D) Plant excretion is impossible, because plants produce no waste that ever needs removing

42. A potted plant is watered until the soil is moist, but a few hours later, on a hot dry windy afternoon, its leaves have wilted while the soil is still moist. Reading the situation, the best explanation is:

- (A) The moist soil water has suddenly turned poisonous and harmed the leaves
- (B) The xylem has transformed into phloem and so stopped carrying water up
- (C) The plant has permanently stopped making its own food in the leaves
- (D) The leaves are losing water by transpiration faster than the roots can supply it from the moist soil

43. Water and dissolved minerals taken up from the soil must reach the leaves at the very top of a tall tree. A student must name the tissue and the direction responsible. The correct pairing is:

- (A) The phloem, carrying water and minerals upward from roots to leaves
- (B) The xylem, carrying food downward from leaves to roots
- (C) The stomata, drawing water sideways around the stem to the top
- (D) The xylem, carrying water and minerals upward from roots to leaves

44. Root hairs are tiny outgrowths on young roots. A student must say which feature makes them so effective at taking up water and minerals. The feature is that they:

- (A) Are green and make their own food to power the absorption of water
- (B) Greatly increase the surface area of the root in contact with soil water
- (C) Contain tiny muscles that actively pump water upward into the stem
- (D) Carry food downward from the leaves into the surrounding soil

45. As water evaporates from the leaves through the stomata, a force is set up that draws a continuous column of water up the stem. A student must name this driving force and state where it originates. The correct answer is:

- (A) A heartbeat in the stem that pumps the water column upward rhythmically
- (B) A phloem push starting at the roots and shoving the water upward
- (C) The transpiration pull, created by water evaporating from the leaves
- (D) A clotting pressure that thickens the sap until it is lifted up the stem

46. Two identical leafy shoots stand in measured water. Shoot 1 is in still, humid shade; shoot 2 is in hot, dry, moving air. Predicting which takes up more water and why, shoot 2 will:

- (A) Take up water more slowly, because heat seals the stomata shut and stops transpiration
- (B) Take up water faster, because hot dry moving air speeds transpiration and so strengthens the pull that draws water up
- (C) Take up exactly the same amount, since outside conditions cannot change transpiration
- (D) Push water back down into the container, reversing the normal upward flow

47. In a young stem the xylem is completely blocked while the phloem stays fully intact. Reasoning about which part suffers FIRST and why, the answer is:

- (A) The leaves, because water can no longer be carried up to them through the blocked xylem
- (B) The leaves, because food can no longer be carried up to them now that the xylem is blocked
- (C) The roots, because the food made above can no longer reach them down the blocked xylem
- (D) The roots, because the minerals they absorbed can no longer leave through the blocked phloem

48. Food made in the leaves must reach a growing root tip below and a ripening fruit above at the SAME time. A student must name the tissue that makes this simultaneous two-way delivery possible. It is the:

- (A) Xylem, which carries food in every direction at once
- (B) Stomata, which push food out to both the fruit and the root
- (C) Phloem, which can move food in more than one direction at once
- (D) Root hairs, which pass food up to the leaves and out to the fruit

49. A ring of bark including the phloem is stripped all the way around a tree trunk, leaving the inner wood (with the xylem) untouched. Over the following weeks, the region that starves and the reason are:

- (A) The trunk and roots below the ring, because food made in the leaves can no longer travel down past the cut phloem to them
- (B) The leaves above the ring, because water can no longer rise to them through the cut
- (C) The whole tree at once, because the ring has severed the xylem that feeds every part
- (D) Only the strip of bark removed, because that is the single part actually cut away

50. A summary row must match each transport tube with its cargo and, where it applies, its direction, across both plant and animal systems. The single fully correct row is:

- (A) Xylem carries food downward; phloem carries water upward; the ureter carries blood
- (B) Xylem carries water and minerals upward; phloem carries food in more than one direction; the ureter carries urine
- (C) Xylem carries water upward; phloem carries water downward; the ureter carries food
- (D) Xylem carries urine; phloem carries minerals upward; the ureter carries water

51. An Amoeba, a single cell, lives without any heart, vessels or xylem, while a tall tree and a human cannot. A student must identify the deciding factor. The deciding factor is that the Amoeba:

- (A) Does not need oxygen, food or water at all to stay alive
- (B) Is so small that materials can enter and leave directly across its surface by diffusion
- (C) Keeps its blood inside one very large single cell instead of in vessels
- (D) Is too simple a creature to require any movement of materials whatsoever

52. A large tree and a human both need transport systems, while a microscopic organism does not. A student must state the underlying reason a big multicellular body needs transport. The reason is:

- (A) It has no outer surface across which any substance could ever pass
- (B) Its inner cells lie too far from the surface to be supplied by simple diffusion alone
- (C) It never requires oxygen, food or water of any kind to live
- (D) It can only survive while it is completely surrounded by water

53. Both the human kidney and a plant's transpiration centrally involve the movement of one particular substance. Naming that shared substance correctly:

- | | |
|---------------|-----------------|
| (A) Water | (B) Haemoglobin |
| (C) Platelets | (D) Bone |

54. A student builds a comparison table of animal and plant transport and must pick the single most reasonable analogy. The most reasonable analogy is:

- (A) The plant's xylem works just like the veins returning blood to the heart
- (B) The plant's phloem works just like the kidneys filtering out the body's waste

- (C) A plant's root hairs act just like the heart's pumping ventricles
- (D) The heart pumps blood through vessels in animals much as the transpiration pull draws water up the xylem in plants, except that plants have no pump

55. A wall chart of transport in living things is checked for the single fully correct summary row. That row is:

- (A) In animals the heart pumps blood through vessels; in plants the xylem carries water up and the phloem carries food
- (B) In animals the xylem pumps blood; in plants the heart carries water to the leaves
- (C) In animals the veins carry blood away from the heart; in plants the phloem carries water up
- (D) Both animals and plants use a heart to move all of their materials around

56. A model of the circulation uses a pump, thick stiff pipes, a fine mesh of one-cell-thin tubing, and floppy valved return pipes. Mapping each model part to the real structure, the fine mesh of thin tubing and the floppy valved return pipes represent, respectively:

- (A) The arteries carrying blood at high pressure, and the capillaries where exchange happens
- (B) The veins returning blood, and the arteries carrying blood away at high pressure
- (C) The capillaries, where exchange with body cells takes place, and the veins, which return blood to the heart through valves
- (D) The heart chambers that collect blood, and the capillaries that exchange with cells

57. A gardener wants to slow water loss from a leafy plant during a heatwave without harming it. Reasoning from how water moves and is lost in plants, the change most likely to reduce water loss is to:

- (A) Place a fan blowing warm dry air across the leaves to help the plant breathe more easily
- (B) Block the root hairs so the plant can no longer pull up any water from the soil
- (C) Move the plant into still, humid shade, since cooler humid still air slows transpiration from the leaves
- (D) Seal the xylem so that water can no longer be carried up to the leaves at all

58. A student must decide which single property belongs to arteries but is NOT typical of veins. The property unique to arteries here is:

- (A) Valves along their length that prevent blood from flowing backward
- (B) A wall that is only a single cell thick to allow exchange
- (C) The job of carrying blood back toward the heart
- (D) Thick, muscular, elastic walls built to withstand high pressure

59. Considering the whole circulation, a student must explain why blood must keep moving continuously rather than being delivered once and left in the tissues. The soundest reason is:

- (A) Because still blood would turn from red to blue inside the body if it stopped moving
- (B) Because the heart's valves would otherwise stop making any sound at all
- (C) Because moving blood is the only thing that prevents it from turning into urine
- (D) Cells constantly use up oxygen and food and keep producing waste, so blood must keep bringing supplies and carrying waste away

60. A doctor finds a patient's wrist pulse and the lub-dub counted with a stethoscope give the same number per minute. A student must explain why these two counts agree. The reason is:

- (A) The pulse is made by the veins while the sound is made by the arteries, and they happen to agree by chance
- (B) Sound travels faster than blood, so the two numbers only appear equal but are not really
- (C) Each heartbeat both makes the valve sounds and sends one surge into the arteries felt as one pulse, so the two counts match
- (D) The stethoscope counts breaths while the wrist counts beats, which happen to coincide

Solutions — Science (Class 7), Chapter 11: Transportation in Animals and Plants — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	C	21	C	31	A	41	A	51	B
2	D	12	A	22	D	32	B	42	D	52	B
3	D	13	A	23	C	33	C	43	D	53	A
4	D	14	B	24	D	34	B	44	B	54	D
5	A	15	B	25	A	35	C	45	C	55	A
6	D	16	B	26	A	36	B	46	B	56	C
7	C	17	C	27	C	37	D	47	A	57	C
8	A	18	B	28	D	38	A	48	C	58	D
9	B	19	A	29	D	39	C	49	A	59	D
10	B	20	A	30	B	40	D	50	B	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Three functions are tested: oxygen carriage (normal), clotting (normal), defence (failing). Only the defence role is impaired, and that role belongs to white blood cells, so they are the deficient component. Red cells carry oxygen, which the chart says is normal, so anaemia is not indicated. Platelets and clotting are explicitly normal here, so a clotting fault is ruled out. Plasma is the liquid carrier; it does not engulf bacteria — that is the white cells' job — so blaming plasma misnames the defender.

2. (D) The packed lower layer is the cells; the yellow upper layer is plasma. Cell fractions are R 60 percent, P 45 percent, Q 40 percent, giving the order R, P, Q. The distractor that ranks by yellow column mistakes plasma for cells. There is no rule that a balanced split carries the most cells. Haemoglobin sits inside the red cells in the packed layer, not dissolved in the yellow plasma, so a tall yellow column signals more plasma, not more cells.

3. (D) A nucleus-free red cell has more space for haemoglobin, raising oxygen carriage; that is the trade-off in the correct option. Losing the nucleus means the red cell cannot divide, so the claim it divides faster is backwards. White cells fight germs, not clotting (that is platelets), so the second wrong option misassigns the job. Haemoglobin belongs to red cells; a white cell losing its nucleus would not start carrying oxygen, so the last option is unfounded.

4. (D) Low iron means too little haemoglobin, reducing oxygen carriage and causing the breathlessness of Patient 1; iron is the dietary link. Slow clotting from small cuts points to too few platelets in Patient 2. Swapping these reverses cause and effect. White cells handle germ defence, not oxygen carriage or clotting, so blaming them for both symptoms is wrong. Plasma volume and red-cell count are not what control oxygen carriage and clot speed here, so the last option misattributes both.

5. (A) Iron is a component of haemoglobin, the oxygen-carrying pigment in red cells, so iron raises oxygen carriage — the correct functional link. Clotting is a platelet job and does not depend on iron, so the platelet claim is wrong. Plasma is mostly water with dissolved substances, not built from iron. White cells are not made of iron, and germ defence does not hinge on it, so the last option is false.

6. (D) White cells are the body's defenders, so their number rises to meet an infection and falls when it is over; red cells, with a different job, stay steady — exactly the observed pattern. Red cells do not turn into white cells; the cell types are distinct. Platelets clot blood and do not convert into white cells. Plasma is the liquid medium and does not solidify into cells, so the last option is biologically impossible.

7. (C) Plasma carries dissolved substances such as digested food and the waste urea around the body — the correct list. Oxygen is carried chiefly by haemoglobin in red cells, not bound to plasma, so the first wrong option overstates plasma's gas role. A clot forms at a wound and is not shuttled around whole. Plasma carries dissolved materials, not whole undigested food chunks or bone fragments, so the last option is wrong.

8. (A) The goal is oxygen delivery, and red cells with their haemoglobin carry oxygen, so they are the directly responsible component. Platelets handle clotting, which is not the stated aim. White cells defend against germs; that is unrelated to oxygen transport. Plasma is the liquid carrier of dissolved substances, but oxygen is carried by haemoglobin in red cells, so plasma proteins are not the answer.

9. (B) Blood leaving the lungs has just picked up oxygen, and oxygen-rich haemoglobin is bright red, so that sample is brighter — the correct reading. Platelet count does not set blood colour, so the entering-sample claim fails. Blood is not blue inside the body; all of it is red, brighter when oxygen-rich, so the 'only lung blood is red' idea is a misconception. Oxygen loading at the lungs does change the colour, so the equal-brightness option is wrong.

10. (B) Red cells are by far the most numerous, are nucleus-free and carry oxygen; white cells are rarer, keep a nucleus and fight germs — the correct pairing. The first wrong option swaps the jobs and names. Platelets do not carry oxygen, so the second wrong option fails. The last option keeps the right names but swaps their jobs, assigning germ-fighting to red cells and oxygen carriage to white cells.

11. (C) Nucleus-free discs are red cells that carry oxygen; the rare nucleated cells are white cells that fight germs. Swapping these two jobs (first wrong option) is the classic flip. Clotting belongs to platelets, and plasma is a liquid not made of cells, so the second wrong option misassigns both. Urea is carried dissolved in plasma, not on red cells, so the last option is wrong.

12. (A) An artery carries blood pumped directly from the heart at high, pulsing pressure, so it spurts with each beat — the correct synthesis. Blood returning slowly to the heart describes a vein and would not spurt. Valves are a vein feature and do not create arterial spurts. Arteries actually have thick walls, not thin ones, so the wall-thickness claim is backwards.

13. (A) Thick muscular walls with pulsing flow and no valves mark arteries, so V1 and V4 are both arteries — V4 is the artery to the lungs, which still pulses even though it carries oxygen-poor blood. Thin walls with valves and low flow mark a vein (V2). A wall one cell thick with the slowest flow is a capillary (V3). The wrong options misname the thin-thick-and-valve pattern or wrongly demote the pulmonary artery to a vein because of its blood content rather than its structure and direction.

14. (B) Blood in veins is at low pressure, and valves prevent backflow so blood keeps moving toward the heart — both corrections needed. The first wrong option repeats the false high-pressure premise. Valves do not produce the heart sounds (those are the heart's own valves closing). Vein valves are passive flaps, not active pumps, so the pumping claim is wrong.

15. (B) Blood leaves the heart through an artery, passes through capillaries where it exchanges oxygen with the muscle, and returns through a vein, giving artery then capillaries then vein. Starting with a vein reverses the trip. Capillaries are the middle exchange step, never first. Veins follow the capillary exchange, so artery-vein-capillaries puts the exchange step last, which is wrong.

16. (B) The rule is a useful generalisation, but the vessel to the lungs (an artery) carries oxygen-poor blood and the vessel back from the lungs (a vein) carries oxygen-rich blood — the named exception. Claiming no exception ignores the lung circuit. Saying arteries always carry oxygen-poor blood reverses the usual case. The rule does not depend on age, so the child-reversal claim is invented.

17. (C) Arteries take blood straight from the pumping heart, so pressure is highest there; capillaries slow the flow so substances have time to exchange with cells. Pressure is not highest in veins (low-pressure return) nor in capillaries. Vein valves exist because of low pressure, not because flow is slowest there. Highest pressure and slowest flow do not coincide in the arteries, so the last option is wrong.

18. (B) All blood is red — darker when oxygen-poor — and the bluish look is an optical effect of light through skin and vessel walls, the correct account. Blood does not start blue and redden in air. Oxygen-poor blood is dark red, not blue. Plasma is pale yellow, not blue, so it cannot tint the vessels blue.

19. (A) Capillary exchange depends on the wall being a single cell thick; thickening it would hinder exchange — the correct consequence. A thicker wall does not push substances out faster; it blocks diffusion. A muscular wall does not turn a vessel into a pump. Thick walls do not trigger clotting, so the clotting claim is unfounded.

20. (A) Carrying blood toward the heart, having valves, and oozing at low pressure all describe a vein; its valves prevent backflow of low-pressure blood — the correct combined statement. Arteries carry blood away from the heart and lack such valves. Capillaries have no valves and are exchange vessels. The last option correctly names a vein but wrongly says its valves cope with high pressure; vein blood is at low pressure.

21. (C) Separating oxygen-rich from oxygen-poor blood means the body receives fully oxygenated blood, maximising oxygen delivery — the real advantage. Loudness of the heartbeat is set by valve closure, not by separation. Separation does not speed colour change, which depends on oxygen loading at the lungs. The heart is muscular and must contract to pump, so the no-muscle claim is false.

22. (D) Thick muscular lower chambers pump blood out forcefully through the arteries, while the thinner upper chambers receive blood from the veins — structure matches function. The first wrong option reverses receive and pump. The heart does not store urine; that is the bladder. The pulse is an artery expanding, not a valve in a chamber, so the last option misreads it.

23. (C) The lub-dub is the heart's valves closing during pumping; the wrist throb (pulse) is an artery expanding as a surge of blood enters it — the correct pairing, and both occur once per beat. Capillaries are silent exchange vessels, and the pulse is not a vein-valve snap. The lub-dub is not breathing, and the pulse is not clotting. Heart valves are in the heart, not at the wrist, so the last option mislocates them.

24. (D) The pulse is the rhythmic expansion of an artery as each surge of pumped blood passes through, the correct identification. It is not a vein-valve snap, which is a common misconception. Blood does not clot and re-dissolve each beat. The lungs move air for breathing, not a throb down the arm, so the air option is wrong.

25. (A) Rate per minute is beats divided by seconds times 60: 33 divided by 22 is 1.5 beats per second, times 60 gives 90 beats per minute, which is a little above a typical resting rate. The 66 figure would come from a wrong division. Reporting 33 forgets to scale the 22 seconds up to a full minute. The 110 figure does not match the arithmetic.

26. (A) In 30 seconds the resting heart (70 per minute) beats 35 times and the post-sprint heart (140 per minute) beats 70 times — twice as many — matching the muscles' raised demand for oxygen and food and faster waste clearance. Halving the count reverses the comparison. A higher rate does raise real flow, so the no-change claim is wrong. A 30-second window holds half a minute's beats, not a full minute's, so 140 in 30 seconds is double-counting.

27. (C) A resting rate of 126 is well above the typical 72 to 80 band, so it is much higher than usual. Values above 100 are not the normal resting figure. A larger number means a faster, not slower, heart, so 'lower' is backwards. Human hearts can exceed 100 beats per minute (for example during exercise), so 126 is not impossible.

28. (D) If one beat takes 0.75 seconds, the number of beats in 60 seconds is 60 divided by 0.75, which equals 80 beats per minute. Reading 0.75 itself as the rate ignores the conversion. The 45 figure has no correct basis. Dividing 100 by 0.75 uses the wrong total (a minute is 60 seconds, not 100), giving the wrong 133.

29. (D) A: 20 divided by 15 times 60 equals 80 per minute. B: 26 divided by 20 times 60 equals 78 per minute. So A is slightly faster despite B having a larger raw count over a longer time. The second option mistakes the larger raw count for a higher rate. The counts do not both give 78. Longer counting does not automatically raise the rate, so the last option's reasoning and numbers are wrong.

30. (B) The double loop sends blood to the lungs to reload oxygen before each body trip, so tissues get oxygen-rich blood — the central benefit. The body does not receive oxygen-poor blood directly; that is the point of the lung loop. The heart cannot skip the lung loop, which constantly reloads oxygen. Mixing the two streams would lower delivered oxygen, the opposite of the benefit.

31. (A) Water rises in the xylem by the transpiration pull created at the leaves, so the plant lifts water without any pump — the matching and its limit. Plants have no heart in the roots or anywhere. The phloem carries food, not water, and does not pump water up. Stomata are leaf pores for water vapour and gas exchange, not soil-water suckers, so the last option is wrong.

32. (B) Excretion is the removal of harmful wastes; taking in and digesting food is the opposite — gaining nutrients — so it is not excretion. Removing urea as urine, losing water and salts in sweat, and passing nitrogen waste through the kidneys are all genuine forms of excretion.

33. (C) The kidney's microscopic filtering units are nephrons, and they sit inside the kidneys, not the bladder (which only stores urine). Stomata are leaf pores, not kidney units. Platelets are blood cells that help clotting, not kidney filters, so both name-substitutions are wrong.

34. (B) Urine forms in the kidney, runs down the ureter to the bladder for storage, then leaves through the urethra: kidney, ureter, bladder, urethra. The other orders scramble these — for instance placing the urethra before the bladder, or starting at the bladder, both of which contradict where urine is made and stored.

35. (C) Urea is made in the body cells, carried dissolved in the blood, filtered by the kidneys, and excreted in urine — the correct trace. Xylem and stomata are plant tissues, not part of human excretion. Urea is not made in the bladder nor breathed out by the lungs. Urea is not produced inside urine, and the heart does not store it, so the last trace is wrong.

36. (B) Urea is a harmful nitrogen waste that must be removed, not a stored food, so the claim is wrong. It is not broken down for energy. The kidneys filter urea out rather than stockpiling it as a reserve. Blood is coloured red by haemoglobin, not urea, so the last option is doubly wrong.

37. (D) Sweat and urine both remove water and some wastes, so the two routes balance: more sweating allows less urine and the reverse — they share the task. Heavy sweating does not shut the kidneys off. Sweat does remove some water and salts, so it is a form of excretion, not merely cooling. Neither organ absorbs oxygen through the skin, so the last option is irrelevant.

38. (A) Dialysis replaces the failed kidneys by filtering wastes from the blood. A transfusion replaces lost blood, not kidney filtering. Transpiration is water loss from plant leaves, not a medical machine. Circulation is the body's own blood movement and is unrelated to replacing kidney function, so the last option is wrong.

39. (C) Plants release some gaseous waste through stomata and store other wastes in leaves, bark, gums and resins — the correct picture. Plants have no kidneys or ureters. Wastes are not all routed up the xylem and out the roots; the xylem carries water and minerals upward. Plants do produce wastes, so the no-waste claim is false.

40. (D) Gums and salt-laden leaves that drop are both ways plants store or shed waste. Food transport is the phloem's job and is not what gum represents. Water moving up the trunk is the xylem's role, unrelated to gum or salt deposits. Oxygen release is a gas-exchange process, not the meaning of gum or accumulated leaf salts.

41. (A) Plants make less harmful waste and can store some, so their excretion is slower and simpler than animals' — the accurate comparison. Plants have no kidneys, so they

cannot be faster for that reason. Plants have no heart or waste vessels. Plants do make waste, so excretion is not impossible.

42. (D) Hot, dry, windy air speeds transpiration, so the leaves lose water faster than the roots can replace it even from moist soil, causing wilting. The soil water has not turned poisonous. Xylem does not turn into phloem. The plant has not permanently stopped making food; wilting is a temporary water-balance problem, so the other options are wrong.

43. (D) Xylem carries water and minerals upward from roots to leaves — the correct tissue and direction. Phloem carries food, not water and minerals up. Xylem does not carry food downward; that is the phloem's role. Stomata are leaf pores for vapour and gas, not water-conducting tubes, so the last option is wrong.

44. (B) Root hairs vastly increase the root's surface area in contact with soil water, boosting absorption — the correct feature. Root hairs are not green and do not photosynthesise. They have no muscles and do not pump. They absorb water and minerals; they do not carry food down into the soil, so the last option is wrong.

45. (C) Water lost from the leaves creates the transpiration pull, which draws water up the stem; it originates at the evaporating leaves — the correct answer. Plants have no heartbeat. A phloem push from the roots is not how water rises; the phloem carries food, and the pull comes from the top. Clotting is a blood process, not a plant water-lifting force.

46. (B) Hot, dry, moving air speeds transpiration, strengthening the pull, so shoot 2 takes up water faster. Heat does not seal the stomata; transpiration rises, not stops. Conditions strongly affect transpiration, so the two shoots differ. Transpiration does not reverse the flow and push water back down, so the last option is wrong.

47. (A) Xylem carries water up; blocking it cuts off water to the leaves, which suffer first. Food travels in the phloem (still intact), not the xylem, so 'food can no longer reach the leaves' misnames the cargo. Food reaching the roots is also a phloem job, unaffected here. Minerals move up in the xylem, not out through the phloem, so the last option confuses both tissue and direction.

48. (C) The phloem can carry food in more than one direction at once, supplying both a root tip and a fruit simultaneously. Xylem carries water and minerals upward, not food in all directions. Stomata are leaf pores, not food-transport tubes. Root hairs absorb water and minerals from soil; they do not distribute food, so the last option is wrong.

49. (A) The phloem carries food down from the leaves; cutting it stops food reaching the trunk and roots below, which starve. Water still rises through the intact xylem, so the leaves above do not lose their water supply. The xylem was not cut, so the whole tree does not fail at once for that reason. The starvation is below the ring, not limited to the removed strip, so the last option is wrong.

50. (B) Xylem carries water and minerals up, phloem carries food in multiple directions, and the ureter carries urine — all correct. The first wrong row swaps xylem and phloem cargo and makes the ureter carry blood. The second makes the phloem carry water and the ureter carry food. The third assigns urine to the xylem and water to the ureter, all wrong.

51. (B) An Amoeba is tiny, so diffusion across its surface supplies and removes materials fast enough — no transport system needed. It does need oxygen, food and water like any cell. It has no blood; it relies on diffusion. It does move materials internally; it simply does so by diffusion, so the no-movement claim is wrong.

52. (B) In a large body, inner cells are too distant from the surface for diffusion alone to supply them, so a transport system is needed. The body does have a surface; the problem is distance, not the absence of a surface. Large organisms certainly need oxygen, food and water. Being surrounded by water is not why transport is needed, so the last option is irrelevant.

53. (A) The kidney removes wastes dissolved in water as urine, and transpiration is the loss of water from leaves, so water is the shared substance. Haemoglobin and platelets are blood components found only in animals, not in plant transpiration. Bone is unrelated to either process, so it is wrong.

54. (D) The closest analogy is that the heart moves blood as the transpiration pull moves water up the xylem, while noting plants lack a pump — the correct, carefully qualified match. The xylem moves water up, not blood back like veins. Phloem carries food, not waste like kidneys. Root hairs absorb water and minerals; they do not pump like ventricles, so the last analogy is wrong.

55. (A) Animals use a heart and vessels; plants use xylem for water up and phloem for food — the fully correct summary. Animals have no xylem, and plants have no heart, so the first wrong row mixes systems. Veins carry blood toward the heart (not away), and phloem carries food, not water, so the second row has two errors. Plants have no heart, so the last row is wrong.

56. (C) A fine mesh of thin tubing is the capillaries (exchange vessels); floppy valved return pipes are the veins (returning blood with valves). The thick stiff pipes — not the thin mesh — are the arteries, so the first wrong option misassigns them. Valved return pipes are veins, not arteries, so the second is reversed. Heart chambers are the pump, not the thin mesh, so the last option is wrong.

57. (C) Cool, humid, still air slows transpiration, so moving the plant into shade reduces water loss without harm. A fan of warm dry air would speed transpiration and worsen loss. Blocking the root hairs starves the plant of water and harms it. Sealing the xylem cuts off the water supply entirely and would kill the plant, so it is not a safe way to slow loss.

58. (D) Arteries have thick, muscular, elastic walls to take the high pressure of blood pumped from the heart — a feature veins lack. Backflow-preventing valves are a vein feature, not an artery one. A single-cell-thick wall describes capillaries. Carrying blood toward the heart is the veins' job, not the arteries', so the other options all belong to different vessels.

59. (D) Cells continuously consume oxygen and food and make waste, so blood must keep circulating to resupply and remove waste — the sound reason. Blood does not turn blue when still; it is always red. The point of circulation is supply and waste removal, not keeping the heart sounds going. Blood does not turn into urine; urine is made by the kidneys from filtered waste, so the last option is wrong.

60. (C) One heartbeat produces both the lub-dub (valves closing) and one arterial surge felt as a pulse, so both counts equal the heart rate — not a coincidence. The pulse is an artery expanding, not made by veins. The agreement is not an illusion of sound speed. The stethoscope is counting heartbeats, not breaths, so the breath-count option is wrong.